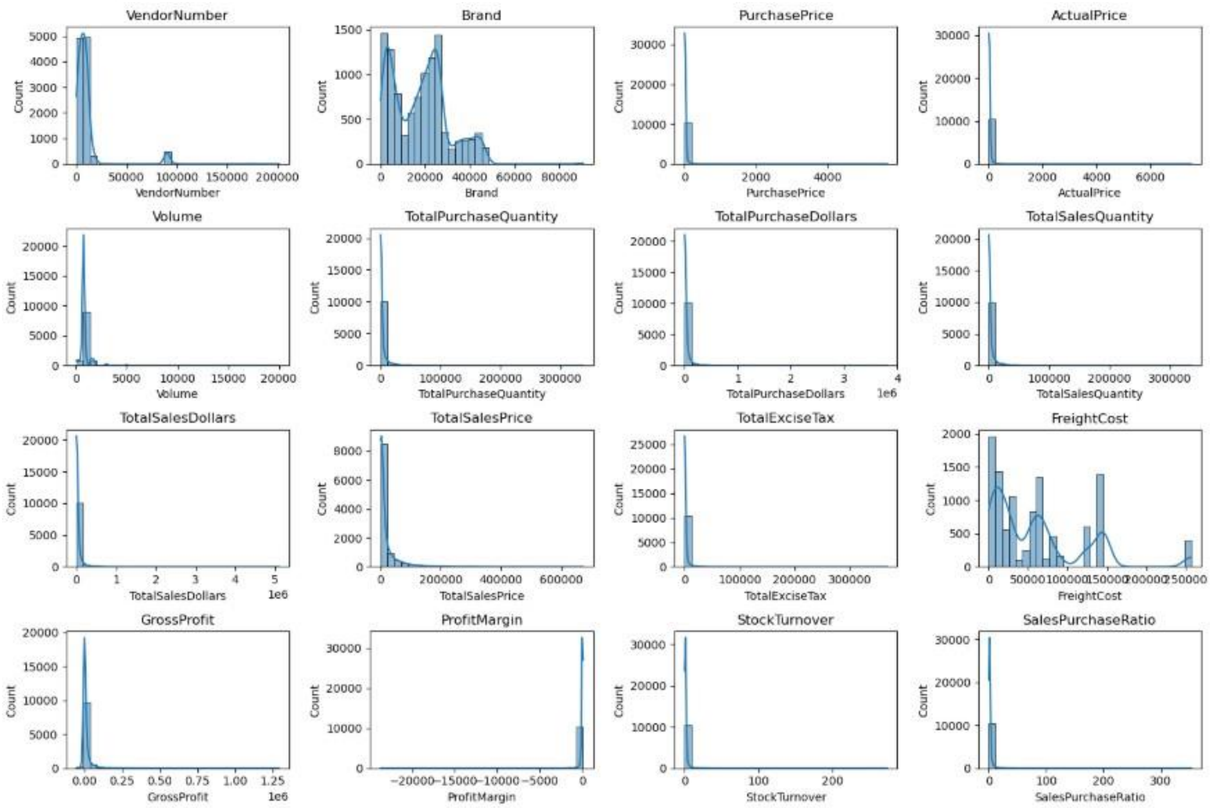
Vendor Sales Performance Report

Problem Statement: Effective inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. Companies need to ensure that they are not incurring losses due to inefficient pricing, poor inventory turnover, or vendor dependency. The goal of this analysis is to:

- Identify underperforming brands that require promotional or pricing adjustments.
- Determine top vendors contributing to sales and gross profit.
- Analyse the impact of bulk purchasing on unit costs.
- Assess inventory turnover to reduce holding costs and improve efficiency.
- Investigate the profitability variance between high-performing and low-performing vendors

Summary Statistics Insights:

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	10650.649458	18753.519148	2.000	3951.00000	7153.0000	9552.00000	201359.000
Brand	10692.0	18039.228769	12662.187074	58.000	5793.50000	18761.5000	25514.25000	90631.000
PurchasePrice	10692.0	24.385303	109.269375	0.360	6.84000	10.4550	19.48250	5681.810
ActualPrice	10692.0	35.643671	148.246016	0.490	10.99000	15.9900	28.99000	7499.990
Volume	10692.0	847.360550	664.309212	50.000	750.00000	750.0000	750.00000	20000.000
TotalPurchaseQuantity	10692.0	3140.886831	11095.086769	1.000	36.00000	262.0000	1975.75000	337660.000
TotalPurchaseDollars	10692.0	30106.693372	123067.799627	0.710	453.45750	3655.4650	20738.24500	3811251.600
TotalSalesQuantity	10692.0	3077.482136	10952.851391	0.000	33.00000	261.0000	1929.25000	334939.000
TotalSalesDollars	10692.0	42239.074419	167655.265984	0.000	729.22000	5298.0450	28396.91500	5101919.510
TotalSalesPrice	10692.0	18793.783627	44952.773386	0.000	289.71000	2857.8000	16059.56250	672819.310
TotalExciseTax	10692.0	1774.226259	10975.582240	0.000	4.80000	46.5700	418.65000	368242.800
FreightCost	10692.0	61433.763214	60938.458032	0.090	14069.87000	50293.6200	79528.99000	257032.070
GrossProfit	10692.0	12132.381048	46224.337964	-52002.780	52.92000	1399.6400	8660.20000	1290667.910
ProfitMargin	10514.0	-15.885224	447.289879	-23730.639	15.35425	30.7785	40.21125	99.717
StockTurnover	10692.0	1.706795	6.020461	0.000	0.80700	0.9820	1.03900	274.500
SalesPurchaseRatio	10692.0	2.504390	8.459071	0.000	1.15400	1.4370	1.66500	352.929



Negative and Zero Values:
Gross Profit: The minimum gross profit recorded is –$52,002.78, indicating that certain products or transactions were sold at a loss. This could be due to high procurement costs or substantial discounts that pushed the selling price below the purchase price.

Profit Margin: The minimum profit margin is negative infinity (–∞). This occurs when revenue is zero or negative, implying that the cost of goods sold exceeded any income generated from the product.

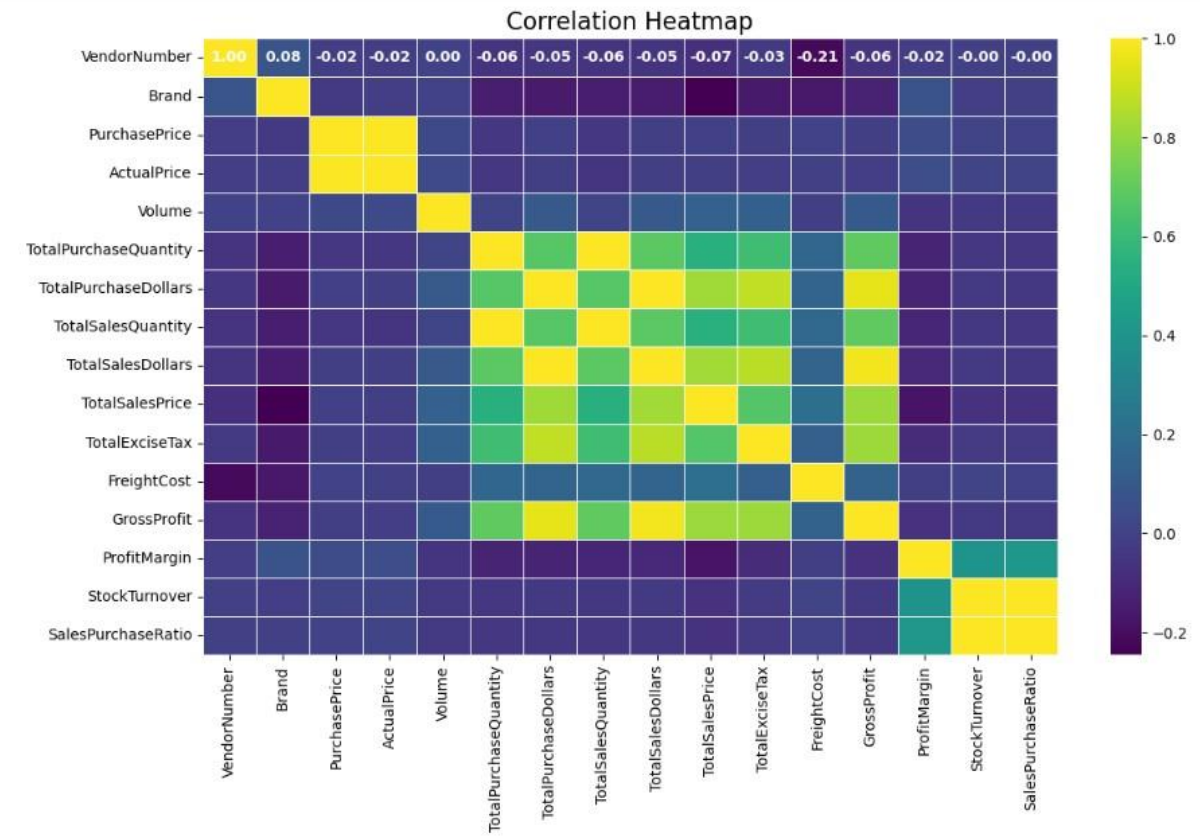
Total Sales Quantity & Sales Dollars: Both metrics show minimum values of zero, suggesting that some purchased products have not yet been sold. These items may represent slow-moving or obsolete stock.

Outliers and High Variability (Reflected by High Standard Deviations):
Purchase & Actual Prices: The maximum purchase price ($5,681.81) and actual price ($7,499.99) are far above their respective averages ($24.39 and $35.64). This points to the presence of a small number of premium, high-value products that skew the distribution.

Freight Cost: Freight charges range from as low as $0.09 to as high as $257,032.07. This wide gap could signal logistical inefficiencies or reflect a combination of small parcel shipments and very large, expensive bulk deliveries.

Stock Turnover: Turnover rates vary from 0 to 274.5. A rate of 0 indicates no sales activity for that inventory, while extremely high rates show exceptionally fast-selling products.
Any turnover above 1 suggests that more items were sold than purchased during the period, meaning sales were fulfilled from existing (older) inventory.

Correlation Insights:



Weak or Insignificant Relationships:

- Purchase Price Impact: The purchase price of a product shows very weak negative correlations with both Total Sales Dollars (-0.012) and Gross Profit (-0.016). This indicates that the cost of an item is not a major driver of its sales revenue or overall profit.

- Stock Turnover vs. Profitability: Stock Turnover has weak negative correlations with Gross Profit (-0.038) and Profit Margin (-0.055). This suggests that simply selling inventory faster does not automatically lead to higher profits.

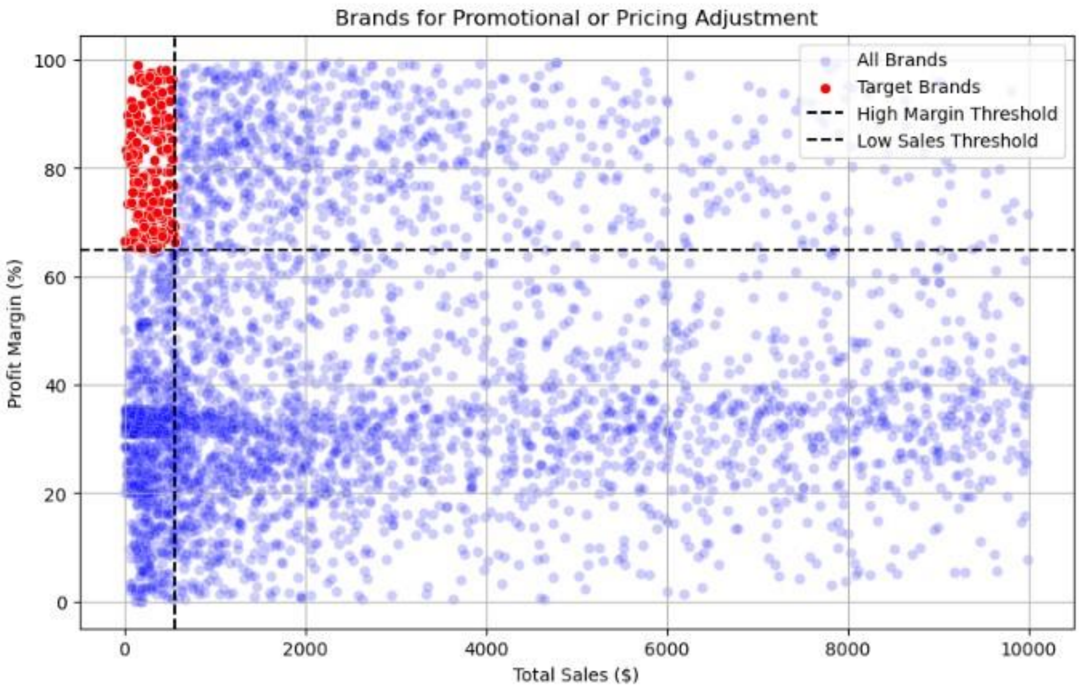
Strong and Notable Relationships:

- Inventory Efficiency: There is a very strong positive correlation (0.999) between the total quantity purchased and the total quantity sold. This confirms highly efficient inventory turnover, meaning most purchased stock is being sold.

- Pricing Pressure: A noticeable negative correlation (-0.179) exists between Profit Margin and the Total Sales Price. This implies that as the selling price increases, the profit margin tends to decrease. A possible explanation is competitive pressure, where higher prices may require increased discounts or incur higher costs, squeezing margins

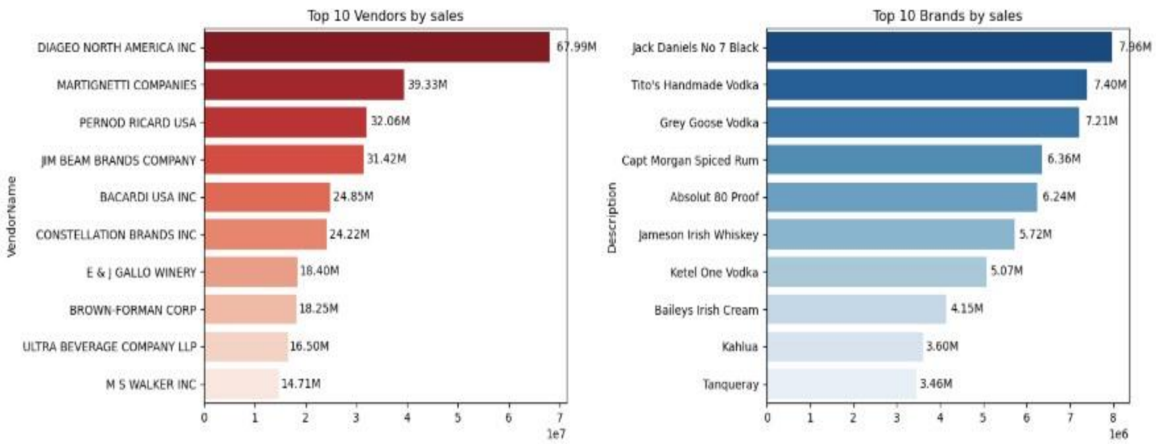
Business Questions:

1. Identify brands that needs Promotional or Pricing adjustments which exhibit lower sales performance but higher profit margin.



Insight: 198 brands are experiencing low sales but maintain high profit margins, positioning them as ideal candidates for focused marketing efforts, promotional initiatives, or strategic pricing changes aimed at increasing sales volume while preserving profitability.

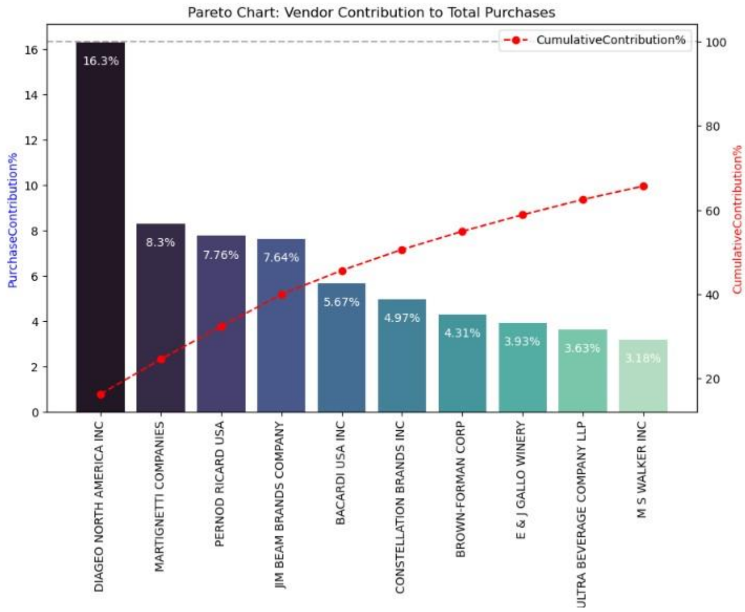
2. Vendors and brands demonstrate the highest sales performance



Insight: The top 10 vendors account for 65.70% of total purchases, while all other vendors together contribute only 34.40%. This heavy dependence on a limited number of vendors increases the risk of supply chain disruptions, highlighting the importance of diversifying the supplier base.

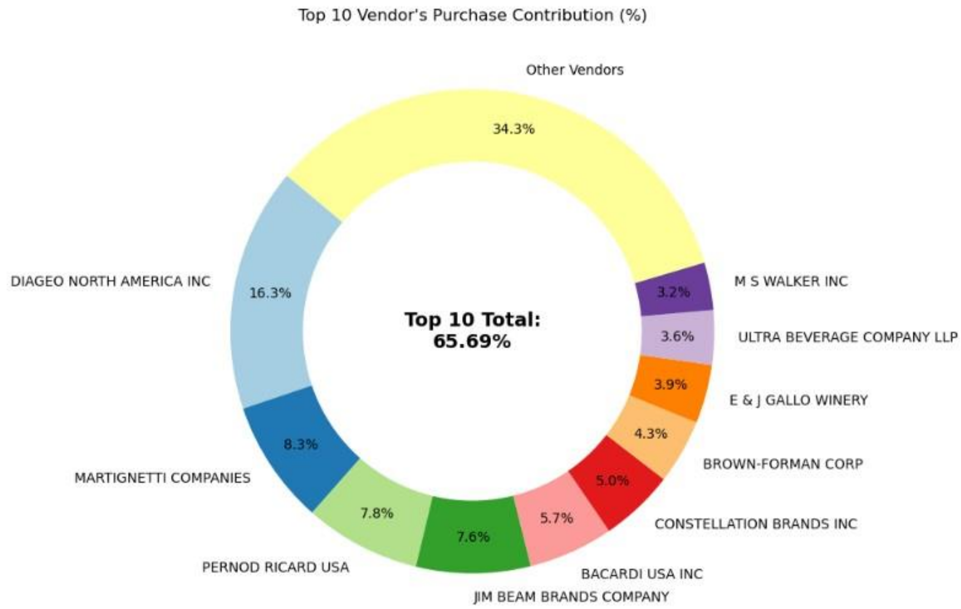
3. Vendors which contribute the most to total purchase dollars

	VendorName	TotalPurchaseDollars
25	DIAGEO NORTH AMERICA INC	50.10M
57	MARTIGNETTI COMPANIES	25.50M
68	PERNOD RICARD USA	23.85M
46	JIM BEAM BRANDS COMPANY	23.49M
6	BACARDI USA INC	17.43M
20	CONSTELLATION BRANDS INC	15.27M
11	BROWN-FORMAN CORP	13.24M
30	E & J GALLO WINERY	12.07M
106	ULTRA BEVERAGE COMPANY LLP	11.17M
53	M S WALKER INC	9.76M



Insight: The top 10 vendors account for 65.7% of total purchases, with Diageo North America Inc. alone contributing 16.3%. This indicates a heavy reliance on a small group of suppliers, which increases risks related to supply chain disruption and pricing power. To reduce vulnerability and strengthen procurement resilience, diversify the supplier base, develop mid-tier vendors

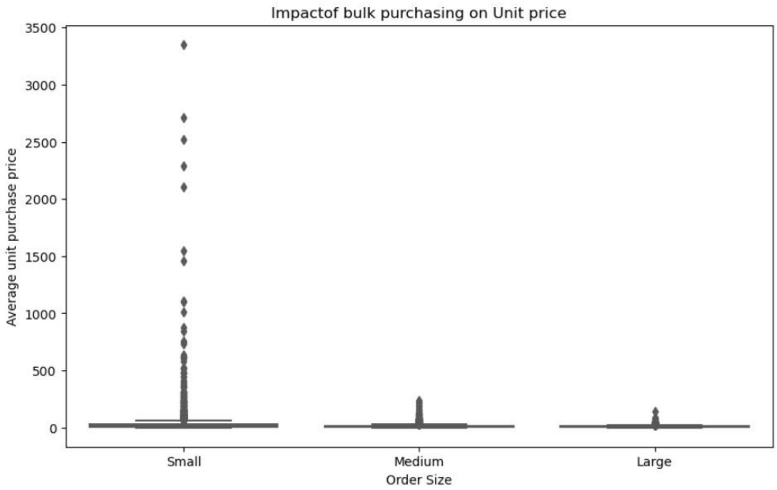
4. Total procurement is dependent on the top vendors:



Insight: The top 10 vendors account for 65.7% of total purchase volume — significantly exceeding the 50% mark. This indicates a potential overdependence on a small group of suppliers. It is essential to diversify our supplier base to reduce risk and improve supply chain resilience.

5. Purchasing in bulk reduces the unit price, and identifying the optimal purchase volume helps maximize cost saving

OrderSize	UnitPurchasePrice
Small	39.068186
Medium	15.486414
Large	10.777625



Insight: Vendors purchasing in larger quantities benefit from a 72% lower unit cost ($10.78 per unit compared to higher costs for smaller orders). Bulk pricing strategies incentivize larger purchases, boosting total sales while preserving profitability.

6. Which vendor have low inventory turnover, indicating excess stock and slow-moving products?

VendorName	StockTurnover
ALISA CARR BEVERAGES	0.615000
HIGHLAND WINE MERCHANTS LLC	0.708000
PARK STREET IMPORTS LLC	0.751000
Circa Wines	0.755692
Dunn Wine Brokers	0.766125
CENTEUR IMPORTS LLC	0.774000
SMOKY QUARTZ DISTILLERY LLC	0.784000
TAMWORTH DISTILLING	0.797000
THE IMPORTED GRAPE LLC	0.807545
WALPOLE MTN VIEW WINERY	0.820667

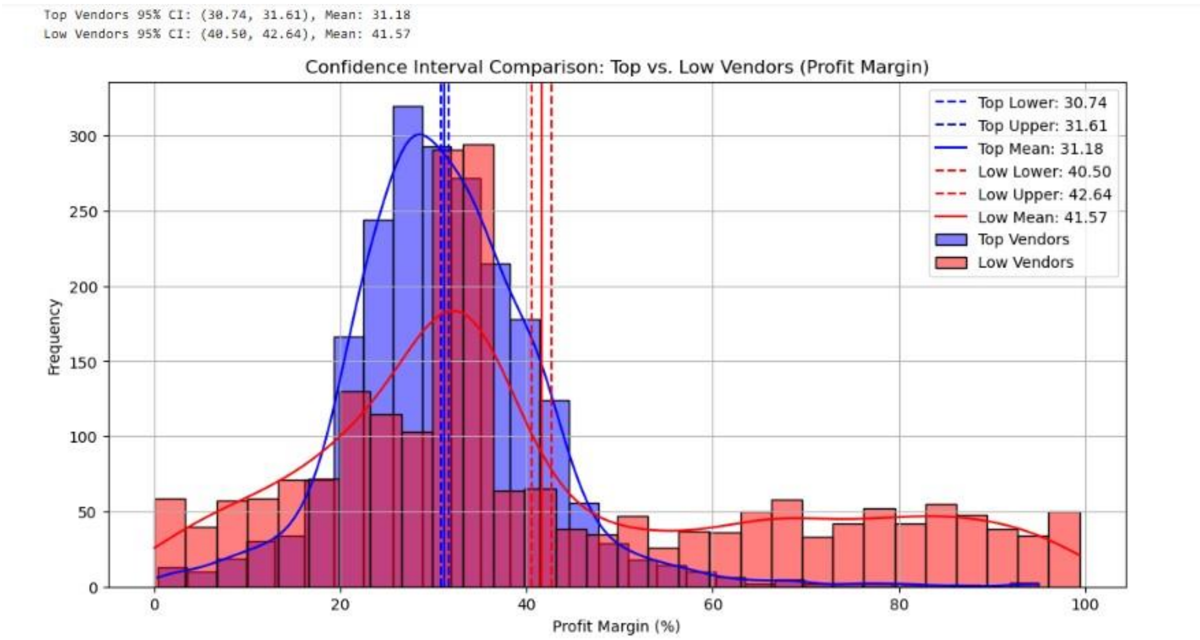
Insight: Slow-moving inventory raises storage costs, limits cash flow efficiency, and negatively impacts overall profitability. Pinpointing vendors with low inventory turnover supports improved stock management and helps reduce financial pressure.

7. How much capital locked in unsold inventory per vendor, and which vendors contribute the most to it?

	VendorName	UnsoldInventoryValue
25	DIAGEO NORTH AMERICA INC	722.21K
46	JIM BEAM BRANDS COMPANY	554.67K
68	PERNOD RICARD USA	470.63K
116	WILLIAM GRANT & SONS INC	401.96K
30	E & J GALLO WINERY	228.28K
79	SAZERAC CO INC	198.44K
11	BROWN-FORMAN CORP	177.73K
20	CONSTELLATION BRANDS INC	133.62K
61	MOET HENNESSY USA INC	126.48K
77	REMY COINTREAU USA INC	118.60K

Insight: A total of over $2.7+ million is locked in unsold inventory across the top 10 vendors, with Diageo, Jim Beam, Pernod Ricard, and William Grant & Sons alone accounting for more than two-thirds of this amount. Immediate action on these vendors like Prioritize sales, liquidation, or return strategies with these top vendors to free up capital and reduce storage costs.

8. What is the 95% confidence intervals for profit margins of top-performing and low-performing vendors?



Insight:

Profit Margin Analysis -

- **Top Vendors:** Profit margin (95% CI): 30.74%–31.61%, Mean: 31.18%
- **Low Vendors:** Profit margin (95% CI): 40.50%–42.64%, Mean: 41.57%
- Low-performing vendors maintain higher margins but experience lower sales volumes, suggesting potential pricing inefficiencies or market-related challenges.

Actionable Insights –

1. Top-Performing Vendors:
Enhance profitability by fine-tuning pricing, reducing operational costs, or offering bundled promotions to increase sales efficiency.

2. Low-Performing Vendors:
Strengthen marketing efforts, adjust pricing strategies, and improve distribution channels to boost sales volume and market reach.

1. Hypothesis Testing: Is there a significant difference in profit margins between top-performing and low-performing vendors?

Null Hypothesis (H0):

- There is no difference in mean profit margin between top and low-performing vendors.

Alternative Hypothesis (H1):

- There is a significant difference in mean profit margin.

Result:

T-Statistics: -17.6695, P-Value: 0.0000

Insights:

- High Sales <> High Profit Margin
- Top-performing vendors (high sales) are achieving lower profit margins.
- Low-performing vendors (low sales) have higher profit margins.

Opportunity Areas –

For high sales / low margin vendors:
- Consider pricing optimization
- Reduce unnecessary discounts
- Negotiate better purchase costs

For low sales / high margin vendors:
- Focus on marketing and distribution
- Increase sales volume without hurting margin

2. Anova Test: Compare the mean of a numeric variable (Profit Margin) across 3 or more groups (Quartiles of vendors, or multiple vendors).

Null Hypothesis (H0):

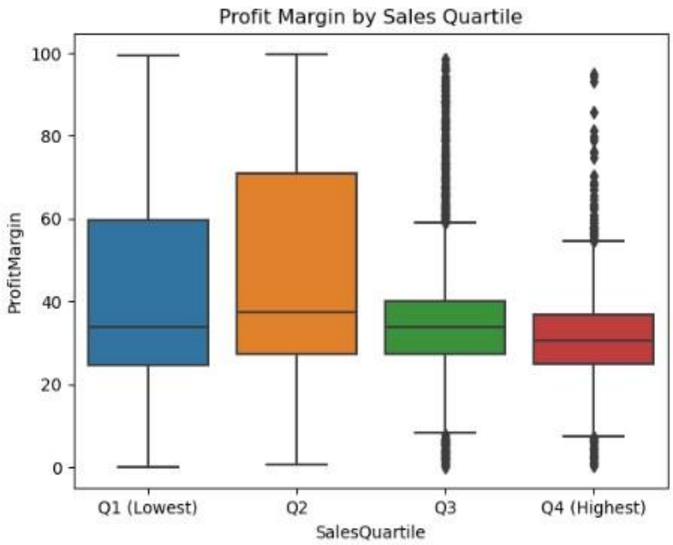
- All quartile means of profit margin are equal.

Alternative Hypothesis (H1):

- At least one quartile profit margin mean is different.

Result:

- F-statistic: 231.8740, p-value: 0.0000

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     Multiple Comparison of Means - Tukey HSD, FWER=0.05
=================================================================
   group1      group2    meandiff p-adj  lower    upper   reject
-----------------------------------------------------------------
Q1 (Lowest)          Q2   5.0375   0.0    3.423   6.6521   True
Q1 (Lowest)          Q3   -6.031   0.0  -7.6455  -4.4164   True
Q1 (Lowest) Q4 (Highest) -10.3913  0.0 -12.0058  -8.7767   True
         Q2          Q3 -11.0685   0.0 -12.6831  -9.4539   True
         Q2 Q4 (Highest) -15.4288  0.0 -17.0434 -13.8142   True
         Q3 Q4 (Highest)  -4.3603  0.0  -5.9749  -2.7457   True
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Finding: Profit margins differ significantly across sales quartiles.

Details from Anova-Tukey HSD:

- Q1 (Lowest Sales) → highest profit margin
- Q4 (Highest Sales) → lowest profit margin ** All quartile pairs differ significantly in profit margin.

Insight:

- Vendors with low sales are often making more profit per unit.
- Vendors with high sales are selling more volume but at lower profit margins.
- Action: consider pricing strategies or cost optimization for Q4 vendors, and marketing push for Q1 vendors.

3. CHI-SQUARE TEST: Is there a significant association between Sales category and Profit Margin category.

Null Hypothesis (H0):

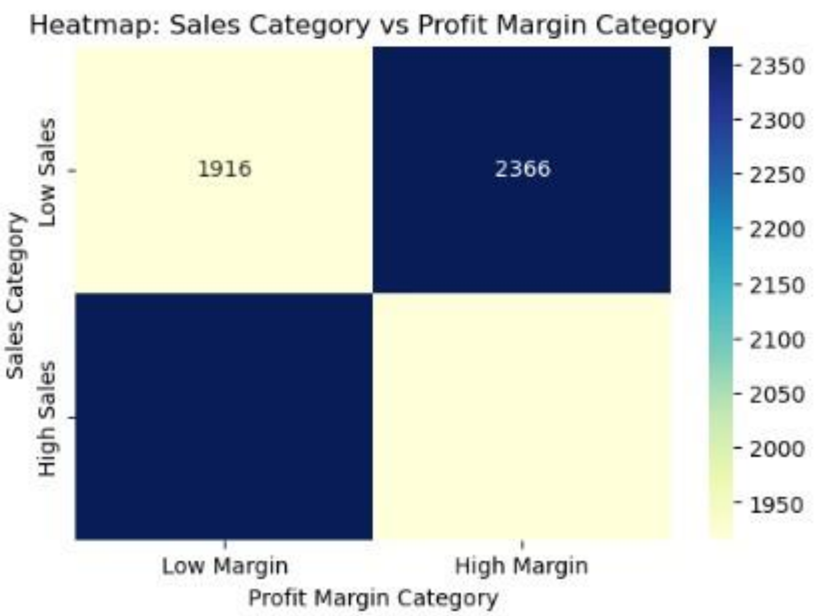
- The sales category (Low Sales / High Sales) and profit margin category (Low Margin / High Margin) are independent.
- There is no association between sales level and profit margin.

Alternative Hypothesis (H1):

- The sales category and profit margin category are not independent.
- There is a significant association between sales level and profit margin

Result:

Chi2: 94.1621, p-value: 0.0000
Reject H0: Sales category and margin category are NOT independent.
There is a significant association between sales and profit margin.



Finding: Chi-square test shows a significant association between sales category and profit margin category (p-value = 0.000).

Heatmap Insight:

- High Sales + Low Margin → Most top-selling vendors have low profit margins → profit is being sacrificed for volume.
- Low Sales + High Margin → Many low-selling vendors have high profit margins → opportunity to increase volume.

Actionable Insight:

- Focus on high sales + low margin vendors to improve profitability through cost reduction or price adjustments.

- Focus on low sales + high margin vendors to boost sales volume through marketing or distribution.

Final Recommendations:

1. **Adjust pricing strategies** for brands with low sales but high margins to increase sales volume while maintaining profitability.
2. **Expand and diversify supplier relationships** to reduce overreliance on a few vendors and minimize supply chain vulnerabilities.
3. **Capitalize on bulk purchasing benefits** to sustain competitive pricing and improve inventory management efficiency.
4. **Manage slow-moving inventory more effectively** by optimizing purchase quantities, introducing clearance promotions, or refining storage practices.
5. **Strengthen marketing and distribution efforts** for underperforming vendors to boost sales volumes without eroding profit margins.
6. **Implementing these actions** will enable the company to drive sustainable profitability, lower risks, and improve overall operational performance.
7. By adopting these **recommendations**, the company can enhance long-term profitability, reduce risks, and streamline its operations for greater efficiency.